

HAMILTON INTL, ON, CanadaWMO: **712630**Lat: **43.1734N**Lon: **79.9340W**Elev: **238**StdP: **98.50**Time Zone: **-5.00 (NAE)**Period: **99-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.7	-15.9	-24.0	0.4	-18.3	-20.7	0.6	-15.4	13.6	-3.0	12.8	-3.2	3.9	240	0.590

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	30.5	22.5	29.0	21.7	27.5	21.0	23.9	28.3	22.9	27.2	22.0	25.9	5.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.6	17.8	26.2	21.6	16.7	25.2	20.8	15.9	24.3	73.1	28.5	69.1	27.3	65.7	26.0	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.2	10.5	9.3	-22.5	32.7	3.4	1.8	-24.9	34.0	-26.9	35.1	-28.8	36.1	-31.2	37.4		
			-22.9	25.7	3.2	1.3	-25.2	26.6	-27.1	27.3	-28.9	28.0	-31.3	28.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.4	-5.2	-4.5	0.1	6.6	13.3	18.5	21.2	20.3	16.8	10.1	4.0	-1.7
	DBStd	10.45	6.28	5.68	5.61	4.83	4.62	3.65	2.94	2.84	3.90	4.59	4.69	5.11
	HDD10.0	1938	471	407	310	125	18	0	0	0	1	57	186	364
	HDD18.3	3902	728	640	566	353	168	41	7	13	75	259	429	622
	CDD10.0	1338	0	0	3	22	121	256	347	320	204	59	6	1
	CDD18.3	261	0	0	0	1	13	47	96	74	28	2	0	0
	CDH23.3	1975	0	0	1	9	129	386	745	499	191	14	0	0
	CDH26.7	489	0	0	0	0	25	105	207	108	43	1	0	0
Wind	WSAvg	4.4	5.1	5.1	4.9	5.1	4.3	3.8	3.4	3.3	3.6	4.3	4.6	5.0
Precipitation	PrecAvg	877	71	58	64	82	80	72	86	71	78	78	71	66
	PrecMax	1130	131	115	139	156	154	194	176	160	193	167	132	151
	PrecMin	641	34	25	23	36	25	20	17	33	21	17	17	28
	PrecStd	126	28	23	32	30	33	40	42	31	35	35	28	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.3	12.4	19.3	24.7	29.3	31.4	32.7	31.9	30.5	25.0	17.7	12.8
		MCWB	11.1	9.2	14.1	17.1	21.1	22.3	23.8	23.6	22.8	19.6	13.2	11.1
	2%	DB	8.6	8.3	14.2	21.1	26.9	29.6	30.9	29.3	27.7	22.0	15.2	9.7
		MCWB	7.5	6.6	10.4	14.4	19.5	21.5	22.9	22.4	21.7	17.5	12.4	8.2
	5%	DB	5.2	5.3	11.0	17.9	24.4	27.6	29.0	27.6	25.5	19.4	13.4	7.1
		MCWB	3.9	3.6	8.2	12.5	17.9	20.9	21.6	21.5	20.0	16.0	10.8	5.8
	10%	DB	2.8	2.7	8.1	14.8	21.7	25.4	27.3	26.1	23.4	17.3	11.2	4.8
		MCWB	1.7	1.3	5.6	10.4	16.2	19.3	20.8	20.7	19.0	14.4	9.1	3.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	10.4	14.8	18.0	22.3	23.7	25.6	25.0	24.1	20.7	14.7	11.2
		MCDB	12.2	11.2	18.0	23.8	27.9	28.9	30.4	28.5	28.4	23.9	16.2	12.2
	2%	WB	7.6	6.7	11.4	15.4	20.5	22.5	24.2	23.6	22.5	18.3	12.9	8.5
		MCDB	8.3	8.2	13.6	19.6	25.6	27.9	28.5	27.7	26.1	20.6	14.9	9.4
	5%	WB	4.0	3.6	8.4	13.3	18.9	21.5	23.0	22.5	21.2	16.4	11.1	5.9
		MCDB	5.0	4.9	10.7	17.0	23.1	26.4	27.4	26.1	24.1	18.9	13.1	7.0
	10%	WB	1.9	1.3	5.6	10.6	17.1	20.3	22.0	21.6	19.9	14.9	9.3	3.7
		MCDB	2.7	2.4	7.8	14.2	20.8	24.2	25.6	24.8	22.6	16.7	11.1	4.7
Mean Daily Temperature Range	5% DB	MDBR	7.2	7.6	8.2	9.6	10.6	10.1	10.4	9.9	9.9	8.6	7.3	6.2
		MCDBR	8.7	10.2	12.6	15.0	13.9	12.5	12.0	11.5	12.2	11.4	10.4	8.3
		MCWBR	8.2	9.0	9.4	10.0	8.2	6.7	6.0	6.0	6.7	7.3	8.2	7.5
	5% WB	MCDBR	8.2	9.8	12.2	13.6	12.7	11.4	10.7	10.0	10.9	10.3	9.6	8.3
		MCWBR	8.1	9.0	9.8	10.2	8.1	6.7	5.9	5.8	6.7	7.3	8.1	7.8
Clear-Sky Solar Irradiance	taub	0.289	0.332	0.330	0.374	0.396	0.432	0.425	0.410	0.377	0.357	0.307	0.291	
	taud	2.401	2.250	2.401	2.314	2.302	2.224	2.266	2.294	2.370	2.413	2.520	2.461	
	Ebn at Noon	848	853	912	891	878	843	845	845	849	821	827	815	
	Edh at Noon	77	106	104	123	128	139	132	124	106	90	69	67	
All-Sky Solar Radiation	RadAvg	1.41	2.16	3.39	4.44	5.42	5.95	5.95	5.23	4.17	2.52	1.57	1.05	
	RadStd	0.14	0.23	0.37	0.48	0.55	0.41	0.43	0.27	0.39	0.23	0.21	0.12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page